

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE FOR ACRYLONITRILE-RICH NITRILE RUBBER SYNTHESIS AND USE

1. ABSTRACT

This report surveys a patent set relevant to high-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-containing elastomer systems, with emphasis on synthesis, composition design, and process control. The evidence shows that the patent landscape is broader than classical NBR manufacture: several patents focus on nitrile-containing copolymers used as binders or dispersants in electrochemical electrodes, while others address controlled emulsion polymerization or block-copolymer architectures that incorporate acrylonitrile or methacrylonitrile segments. Across the set, the target material is defined by elevated nitrile content and/or nitrile-rich copolymer segments, often paired with alkylene, diene, or other hydrophobic structural units to balance polarity, dispersibility, and mechanical performance.

The most directly relevant synthesis-oriented disclosures are those describing nitrile-containing copolymers with controlled monomer ratios and molecular weight windows. EP3920285B1 specifies a binder composition containing a copolymer with 40 to 70 mass% alkylene structural units and 30 to 55 mass% nitrile group-containing monomer units in N-methylpyrrolidone, with Mooney viscosity of 25 or less and iodine value of 8 mg/100 mg or more. US11784313B2 similarly uses hydrogenated nitrile butadiene rubbers with alkylene and nitrile units, but emphasizes a dual-binder strategy with a high-MW first rubber and a lower-MW second rubber for carbon nanotube dispersion and electrode adhesion. EP3900086B1 defines epoxy group-containing nitrile rubber copolymers containing 5 to 60 wt% α,β -unsaturated nitrile units and 10 to 94.99 wt% conjugated diene units, showing a functionalized route for battery binders. US11827830B2 discloses a nitrile-containing copolymer dispersant with 15 to 50 mass% nitrile units and 40 to <85 mass% aliphatic hydrocarbon units, while US7592409B2 demonstrates controlled emulsion polymerization with in-situ emulsifier formation for acrylonitrile-containing block polymers. EP2057205B1 contributes a metathesis-based route to meta-block copolymers from unsaturated polymers, relevant as an alternative architecture platform rather than a direct NBR route. Overall, the landscape favors emulsion or solution-based copolymerization, careful monomer-ratio control, and molecular-weight/viscosity tuning to achieve high nitrile content without sacrificing processability or end-use performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP3428214B1
- Patent Title: Polymer fine particle-containing curable resin composition having improved ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US7592409B2
- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US11827830B2
- Patent Title: Dispersant, conductive material dispersion and slurry for electrode film
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US10392463B2

- Patent Title: Acrylic resin composition, and molded product and film made from same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US8394877B2
- Patent Title: Method for manufacturing porous structure and method for forming pattern
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US11784313B2
- Patent Title: Conductive material paste composition for secondary battery electrode, slurry ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: EP3900086B1
- Patent Title: Electrode composition for a cathode of a cell of a lithium-ion battery, a ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: EP3920285B1
- Patent Title: Binder composition for secondary battery electrode, conductive material paste ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US10629899B1
- Patent Title: Production method for electrochemically stable anode particulates for lithium ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Monomer content windows differ by application: EP3920285B1 uses 30 to 55 mass% nitrile units with 40 to 70 mass% alkylene units; US11827830B2 uses 15 to 50 mass% nitrile units with 40 to <85 mass% aliphatic hydrocarbon units; EP3900086B1 uses 5 to 60 wt% α,β -unsaturated nitrile units with 10 to 94.99 wt% conjugated diene units.
- US11784313B2 and EP3920285B1 both target electrode binders/dispersants, but US11784313B2 emphasizes a two-rubber binder system (high-MW first rubber plus lower-MW second rubber) to improve fibrous carbon nanomaterial dispersion and adhesion, whereas EP3920285B1 focuses on a single copolymer in NMP with low Mooney viscosity and defined iodine value.
- US11827830B2 is the most compositionally broad dispersant disclosure, combining nitrile units with aliphatic hydrocarbon and carbamoyl units; unlike the battery-binder patents, it explicitly allows a carbamoyl-containing structural unit up to 10 mass% and specifies a weight-average molecular weight of 5,000 to 400,000.
- US7592409B2 is process-centric rather than end-use-centric: it discloses emulsion polymerization with in-situ emulsifier formation and controlled free-radical polymerization agents, enabling acrylonitrile/methacrylonitrile-containing block polymers; this contrasts with the more formulation-driven battery binder patents.
- EP2057205B1 differs fundamentally from the nitrile-copolymer patents by using metathesis of unsaturated polymers to make meta-block copolymers; it is relevant as an alternative architecture/control strategy, but it does not define NBR monomer ratios or conventional nitrile-rubber emulsion polymerization conditions.
- Across the set, newer patents trend toward functionalized or hydrogenated nitrile rubbers for electrode applications, while older process disclosures emphasize controlled emulsion polymerization and block-copolymer construction rather than direct high-ACN NBR manufacture.
- No patent in the set provides a complete high-ACN NBR recipe with explicit acrylonitrile/butadiene feed ratio, initiator, emulsifier, chain-transfer agent, temperature, pressure, conversion, and coagulation sequence; the evidence instead supports a design space centered on nitrile-rich copolymers with controlled viscosity and molecular weight.

4. CONCLUSION

The patent evidence supports high-acrylonitrile NBR development through tight control of nitrile content, molecular weight, and polarity-balanced comonomer selection rather than through a single universal synthesis recipe. For practical synthesis, the strongest evidence favors nitrile-rich copolymers in the approximate range of 30 to 55 mass% nitrile units for binder/dispersant use, with alkylene or diene-derived segments providing flexibility and processability. Functionalization

by epoxy groups or partial hydrogenation appears repeatedly as a route to improve compatibility and electrode performance. Process-wise, controlled emulsion polymerization with in-situ emulsifier formation is the clearest disclosed manufacturing approach among the retained patents, while solution-based formulation in NMP and viscosity control via Mooney or molecular-weight tuning are important downstream handling strategies. For high-acrylonitrile NBR specifically, the evidence recommends prioritizing a nitrile-rich copolymer backbone, low-to-moderate Mooney viscosity for dispersion, and careful balancing of hydrophobic backbone content to preserve film formation and adhesion. However, the manifest does not disclose a single patent with a fully specified high-ACN NBR polymerization protocol, so any exact initiator, temperature, pressure, or coagulation conditions would require additional source evidence beyond this set.

5. REFERENCES

- EP3428214B1 | Polymer fine particle-containing curable resin composition having improved ... | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP3428214B1/en>
- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US11827830B2 | Dispersant, conductive material dispersion and slurry for electrode film | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11827830B2/en>
- US10392463B2 | Acrylic resin composition, and molded product and film made from same | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10392463B2/en>
- US8394877B2 | Method for manufacturing porous structure and method for forming pattern | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8394877B2/en>
- US11784313B2 | Conductive material paste composition for secondary battery electrode, slurry ... | Not disclosed | US | 2023 | <https://patents.google.com/patent/US11784313B2/en>
- EP3900086B1 | Electrode composition for a cathode of a cell of a lithium-ion battery, a ... | Not disclosed | EP | 2023 | <https://patents.google.com/patent/EP3900086B1/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- EP3920285B1 | Binder composition for secondary battery electrode, conductive material paste ... | Not disclosed | EP | 2024 | <https://patents.google.com/patent/EP3920285B1/en>
- US10629899B1 | Production method for electrochemically stable anode particulates for lithium ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10629899B1/en>